EXPERIMENT: 14

Simulation of Edge and Cloud Processing for an IoT Application

OBJECTIVES :

1. To understand the concepts of edge computing and cloud computing in IoT applications.

2. To simulate IoT data processing at the edge device and at a remote cloud server using MATLAB.

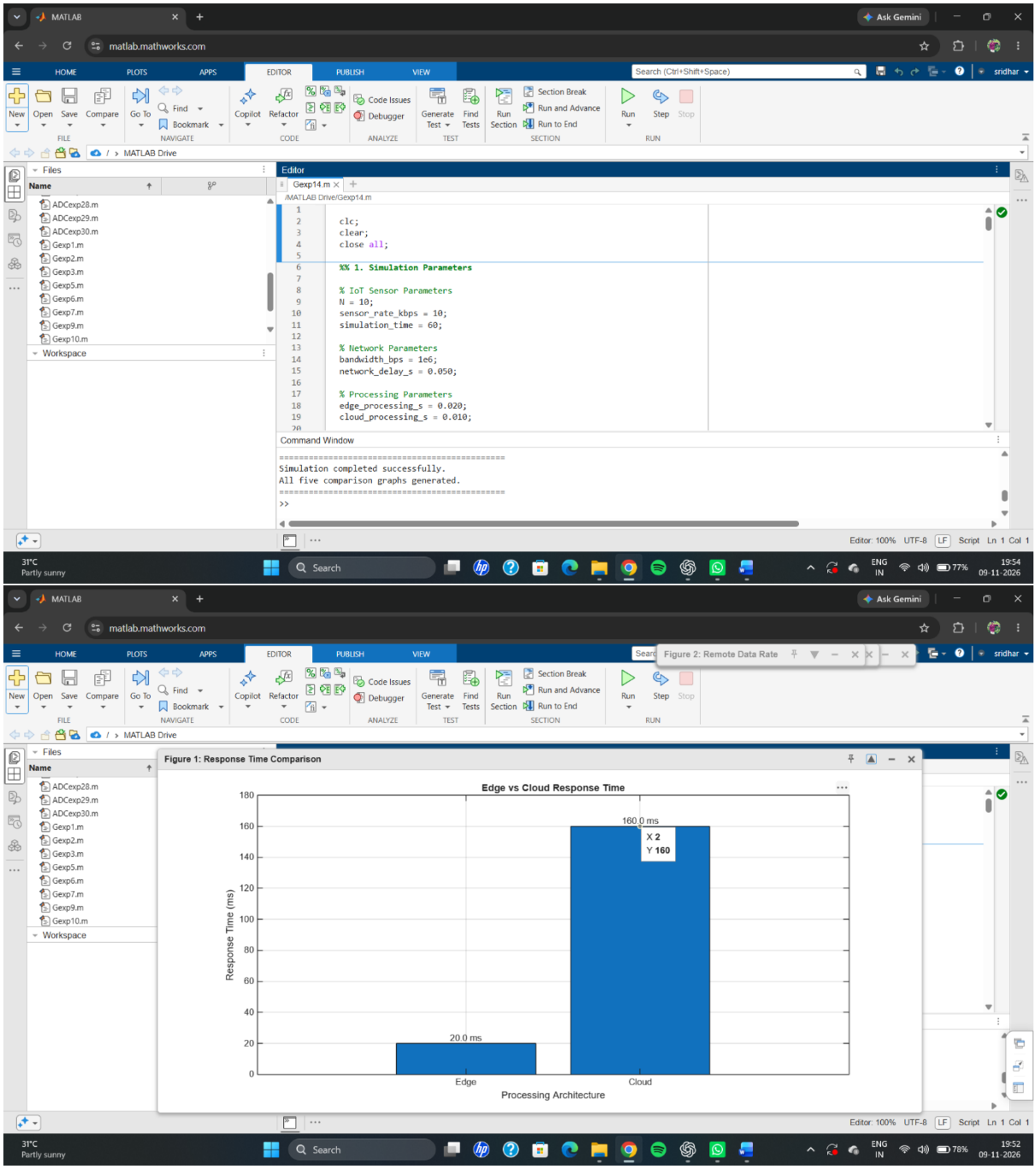
3. To compare edge and cloud processing in terms of latency, network bandwidth utilization, response time, and energy consumption.

4. To analyze the effect of network delay and the volume of sensor data on overall system performance.

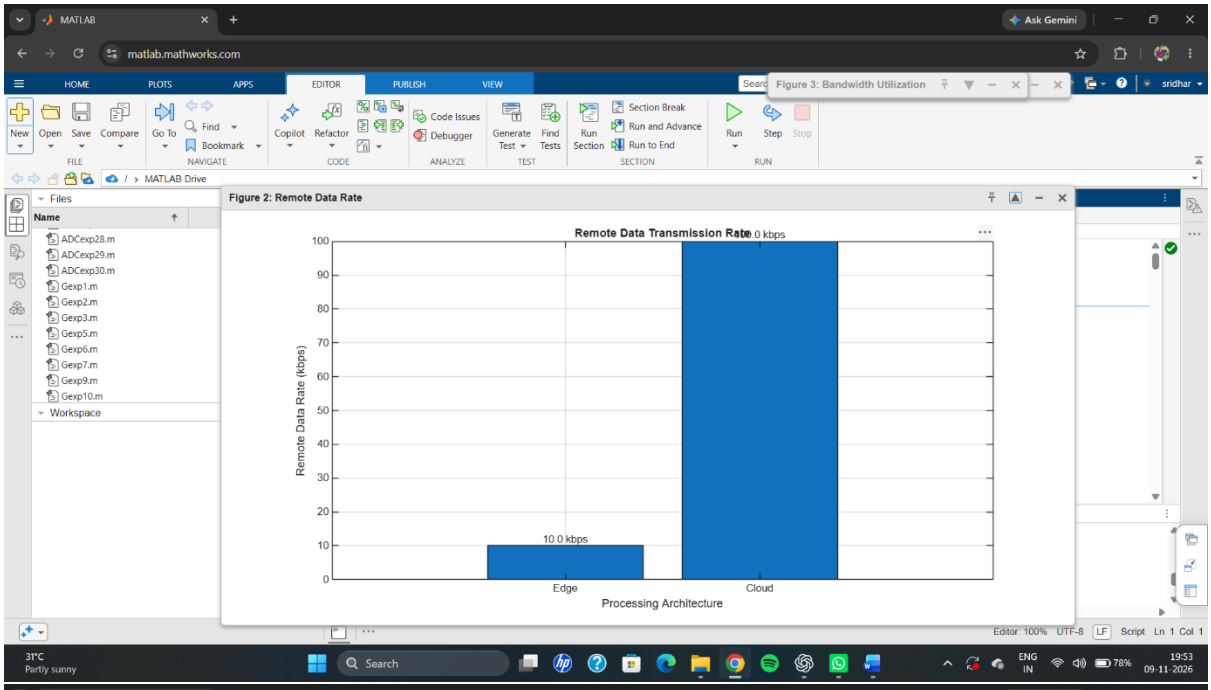
5. To evaluate the suitability of edge and cloud processing for real-time IoT applications.

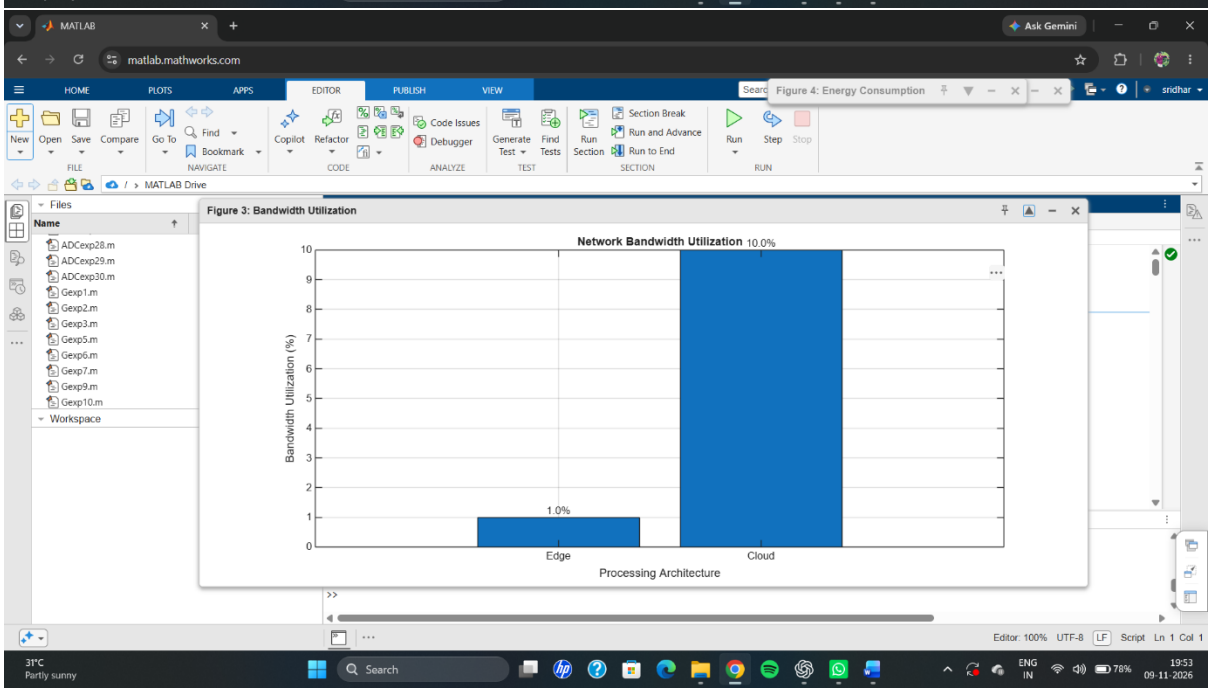
6. To generate comparative plots and identify the advantages and limitations of both processing approaches.

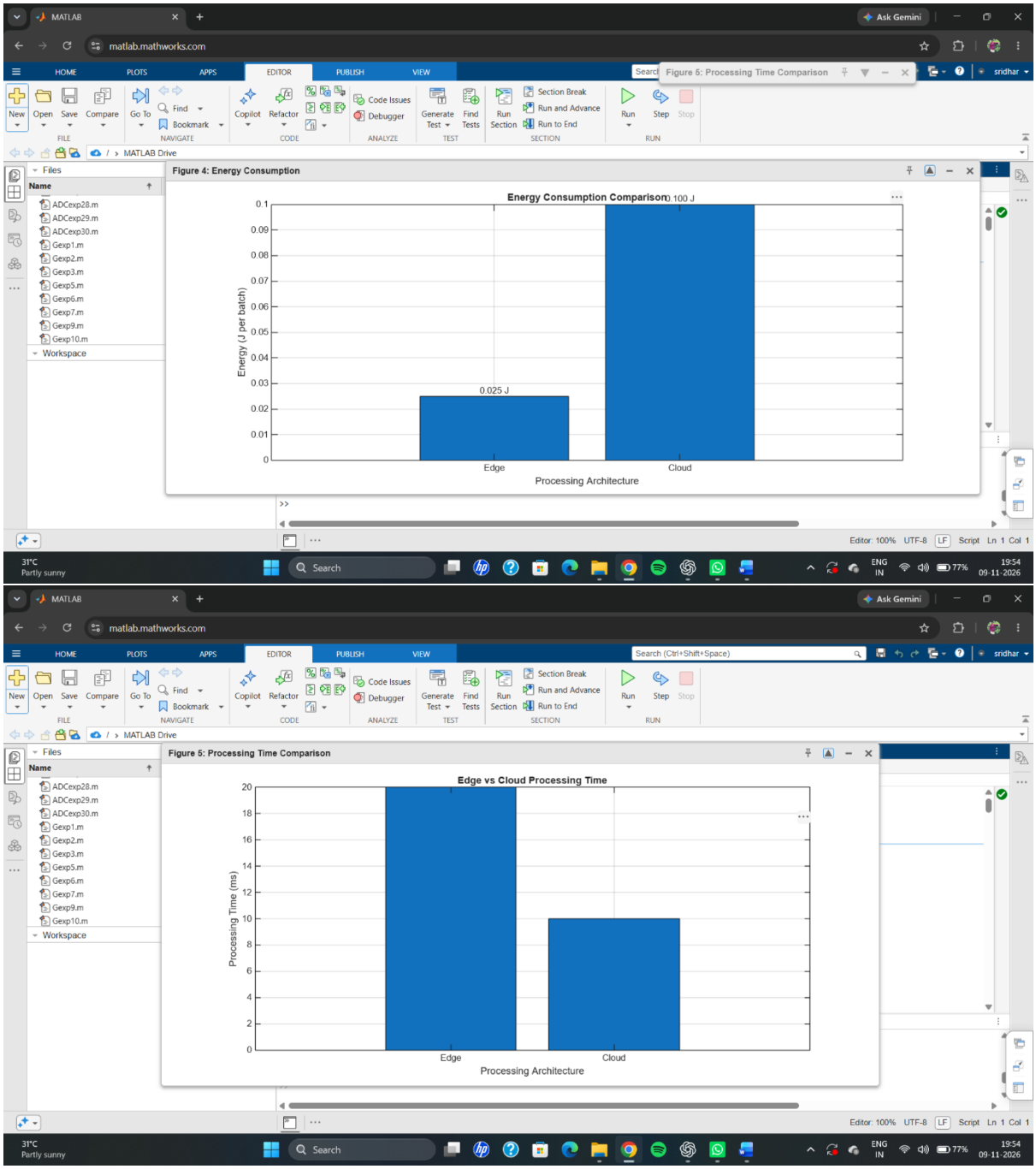
PROGRAME CODE :


MATLAB
Ask Gemini
matlab.mathworks.com
HOME
PLOTS
APPS
EDITOR
PUBLISH
VIEW
sridhar
New
Open
Save
Compare
Go To
Find
Bookmark
Copilot
Refactor
Code Issues
Debugger
Generate Test
Find Tests
Run Section
Section Break
Run and Advance
Run to End
Run
Step
Stop
FILE
NAVIGATE
CODE
ANALYZE
TEST
SECTION
RUN
MATLAB Drive
Files
Name
ADCexp28.m
ADCexp29.m
ADCexp30.m
Gexp1.m
Gexp2.m
Gexp3.m
Gexp5.m
Gexp6.m
Gexp7.m
Gexp9.m
Gexp10.m
Workspace
Editor
Gexp14.m
/MATLAB Drive/Gexp14.m
clc;
clear;
close all;
%% 1. Simulation Parameters
% IoT Sensor Parameters
N = 10;
sensor_rate_kbps = 10;
simulation_time = 60;
% Network Parameters
bandwidth_bps = 1e6;
network_delay_s = 0.050;
% Processing Parameters
edge_processing_s = 0.020;
cloud_processing_s = 0.010;
Command Window
Simulation completed successfully.
All five comparison graphs generated.
>>
Editor: 100% UTF-8 LF Script Ln 1 Col 1
31°C
Partly sunny
Search
ENG
IN
77%
19:54
09-11-2026

MATLAB
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Figure 2: Remote Data Rate
Figure 1: Response Time Comparison
Edge vs Cloud Response Time
Response Time (ms)
Processing Architecture
Edge
Cloud
20.0 ms
160.0 ms
X 2
Y 160
0
20
40
60
80
100
120
140
160
180
Editor: 100% UTF-8 LF Script Ln 1 Col 1
31°C
Partly sunny
Search
ENG
IN
78%
19:52
09-11-2026


MATLAB
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Figure 3: Bandwidth Utilization
Figure 2: Remote Data Rate
Remote Data Transmission Rate 100.0 kbps
Remote Data Rate (kbps)
10.0 kbps
Edge
Cloud
Processing Architecture
MATLAB Drive
Files
Workspace
Search
31°C
Partly sunny
09-11-2026


MATLAB
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matlab.mathworks.com
Figure 4: Energy Consumption
Figure 3: Bandwidth Utilization
Network Bandwidth Utilization 10.0%
Bandwidth Utilization (%)
1.0%
Edge
Cloud
Processing Architecture
MATLAB Drive
Files
Workspace
Search
31°C
Partly sunny
09-11-2026



RESULT : The MATLAB simulation of Edge and Cloud Computing for IoT-based data processing was successfully implemented and analyzed. The results showed that edge computing reduces response time from 160 ms to 20 ms, decreases remote data transmission by 90%, and consumes less energy per batch compared with cloud computing. Hence, edge computing provides better latency, bandwidth efficiency, and energy efficiency for time-sensitive IoT applications under the given simulation parameters.